III.2 Statistika plynů

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III.2 Statistika kvantových plynů

→ grandkanonický model rovnovážného stavu

$$\hat{D}_G \Big|_{\mathcal{H}^\pm_{Fock}} = \frac{1}{Z_G} e^{-\beta(\hat{H} - \mu\hat{N})}$$

partiční suma:

$$Z_G = \mathrm{Tr}_{\mathcal{H}^\pm_{Fock}}\, e^{-\beta\sum_k (E_k-\mu)\hat{n}_k}$$

$$= \sum_{n_0,n_1,\dots} \underbrace{e^{-\beta\sum_k (E_k-\mu)n_k}}_{\prod_k e^{-\beta(E_k-\mu)n_k}} = \prod_k \sum_n \Big(\underbrace{e^{-\beta(E_k-\mu)}}_{X_k}\Big)^n$$

$\oplus$ $\quad n \in \{0,\dots,\infty) \Rightarrow \sum_n X_k^n = \frac{1}{1-X_k}$

$\ominus$ $\quad n \in \{0,1\} \Rightarrow \sum_n X_k^n = 1 + X_k$

$$= \prod_k (1 \mp X_k)^{\mp 1}$$

$$\ln Z_g^\pm = \mp \sum_k \ln\left[1 \mp e^{-\beta(E_k-\mu)}\right]$$

Statistické vlastnosti

Fermiony

→ $\ln Z_g^- = \sum_k \ln\big(1 + e^{-\overbrace{\beta(E_k-\mu)}^{\lambda_k}}\big)$

- $\langle n_k \rangle = -\frac{\partial \ln Z_g^-}{\partial \lambda_k} = \frac{e^{-\lambda_k}}{1+e^{-\lambda_k}} = \frac{1}{1+e^{\beta(E_k-\mu)}} = f_k^-$
- pro fermiony $\langle n_k \rangle \equiv \mathbb{P}(n_k = 1)$
- $\mathrm{Cov}(n_k,n_l) = \frac{\partial^2 \ln Z_g}{\partial\lambda_k \partial\lambda_l} = -\delta_{kl}\frac{\partial \langle n_k\rangle}{\partial \lambda_k} = \delta_{kl}\frac{e^{\lambda_k}}{(1+e^{\lambda_k})^2}$

$$= f_k^-(1-f_k^-)\,\delta_{kl}$$

Bosony

→ $\ln Z_g^+ = -\sum_k \ln(1 - e^{-\lambda_k})$

- $\langle n_k \rangle = -\frac{\partial \ln Z_g^+}{\partial \lambda_k} = \frac{e^{-\lambda_k}}{1-e^{-\lambda_k}} = \frac{1}{e^{\beta(E_k-\mu)}-1} = f_k^+$
- $\mathrm{Cov}(n_k,n_l) = \frac{\partial^2 \ln Z_g^+}{\partial\lambda_k \partial\lambda_l} = -\frac{e^{\lambda_k}}{(e^{\lambda_k}-1)^2}\,\delta_{kl}$

$$= f_k^+(1+f_k^+)\,\delta_{kl}$$

$$\langle n_k \rangle^\pm = f_k^\pm = \frac{1}{e^{\beta(E_k-\mu)} \mp 1}$$

$$\mathrm{Cov}(n_k,n_l) = f_k^\pm(1 \pm f_k^\pm)$$

$$\mathbb{P}(n_k = n) = \begin{cases} f_k^- & \text{Bernoulliho rozdělení} \\ e^{-\lambda_k n_k}(1-e^{-\lambda_k}) & \text{Geometrické rozdělení} \end{cases}$$